

Java Darleen Villano

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Academic positions

University of Toronto <i>Postdoctoral Fellow</i>	2025-2026
University of Connecticut <i>Graduate Student</i>	2019-2025

Education

University of Connecticut <i>Ph.D. Mathematics</i>	2019-2025
◦ Advisers: Reed Solomon and Damir D. Dzhafarov	
◦ Dissertation Title: Computable Categoricity, and Topology in Reverse Mathematics	
University of California, Berkeley <i>B.A. Mathematics with Logic Minor</i>	2015-2019

Research interests

Computability theory, computable structure theory, reverse mathematics, and algorithmic randomness.

Publications

Extensions of categoricity relative to a degree	June 2026
Villano, J.D.	
Accepted, <i>Zeitschrift für Mathematische Logik und Grundlagen der Mathematik</i> 🔗	
Computable categoricity relative to a c.e. degree	November 2025
Villano, J.D.	
<i>Notre Dame Journal of Formal Logic</i> 🔗	
Normality, relativization, and randomness	June 2025
Calvert, W., Gruner, E., Mayordomo, E., Turetsky, D., Villano, J.D.	
<i>Theory of Computing Systems</i> 🔗	
The Ginsburg–Sands theorem and computability theory	May 2024
Benham, H., DeLapo, A., Dzhafarov, D., Solomon, R., Villano, J.D.	
<i>Advances in Mathematics</i> 🔗	

Preprints

Switching Categoricity Behavior with a Given Degree	September 2026
Gonzalez, D., Morales, J.J.V., Villano, J.D.	
arXiv:2609.21047 🔗	

Teaching experience

Primary Instructor	Toronto, ON
<i>University of Toronto</i>	2025-2026
◦ Fall 2025, Winter 2026: MAT133Y (Calculus and Linear Algebra for Commerce), 1 section	
Primary Instructor	Storrs, CT
<i>University of Connecticut</i>	2023-2024
◦ Fall 2024: Math 1071Q (Calculus for Business and Economics), 2 sections	
◦ Spring 2024: Math 1071Q (Calculus for Business and Economics), 1 section	
◦ Fall 2023: Math 1071Q (Calculus for Business and Economics), 2 sections	
◦ Spring 2023: Math 1071Q (Calculus for Business and Economics), 2 sections	

Teacher Assistant
University of Connecticut

Storrs, CT
2019-2022, 2025

- **Spring 2025:** Math 2110Q (Multivariable Calculus), 3 sections
- **Fall 2022:** Math 1132Q (Calculus II), 2 sections
- **Spring 2022:** Math 1132Q (Calculus II), 2 sections
- **Fall 2021:** Math 1131Q (Calculus I), 2 sections
- **Spring 2021:** Math 1132Q (Calculus II), 2 sections
- **Fall 2020:** Math 1132Q (Calculus II), 2 sections
- **Spring 2020:** Math 1132Q (Calculus II), 2 sections
- **Fall 2019:** Math 1131Q (Calculus I), 2 sections

Conference invitations

AMS Fall Eastern Sectional Meeting
George Washington University

Washington, D.C.
Fall 2026

Presentation title: Dichotomy for classes of countable graphs forbidding two finite subgraphs

AMS Fall Central Sectional Meeting
St. Louis University

St. Louis, MO
Fall 2025

Presentation title: Categorical behavior in the Turing degrees

Workshop “Reverse Mathematics: New Paradigms”
Erwin Schrödinger International Institute for Mathematics and Physics
Presentation title: Relativized computable categoricity

Vienna, Austria
Summer 2025

Summer School “Reverse Mathematics: New Paradigms”
Erwin Schrödinger International Institute for Mathematics and Physics

Vienna, Austria
Summer 2025

Logicón 2025
Facultad de Ciencias UNAM

México City, México
Spring 2025

Presentation title: Computable categoricity relative to a degree (online talk)

ASL North American Annual Meeting
New Mexico State University

Las Cruces, NM
Spring 2025

Presentation title: Computable categoricity relative to a generic degree

Dagstuhl Seminar – Weihrauch Complexity: Structuring the Realm of Non-Computability
Schloss Dagstuhl

Wadern, Germany
Spring 2025

South Eastern Logic Symposium
University of Florida

Gainesville, FL
Spring 2025

Presentation title: Computable categoricity relative to a degree

Graduate Research Forum
University of Connecticut

Storrs, CT
Spring 2025

Presentation title: Relativizing computable categoricity

The New England Recursion and Definability Seminar
Dartmouth College

Hanover, NH
Fall 2024

Presentation title: Computable categoricity relative to a c.e. degree

Computable Structure Theory and Interactions
Technische Universität Wien

Vienna, Austria
Summer 2024

Presentation title: Computable categoricity relative to a degree

Joint Mathematics Meeting – AMS Special Session on Computable Mathematics: A Session Dedicated to Martin D. Davis

San Francisco, CA
Spring 2024

Presentation title: Computable categoricity relative to a c.e. degree

A Convergence of Computable Structure Theory, Analysis, and Randomness

Banff, AB
Spring 2023

Banff International Research Station

AMS New England Graduate Student Conference

Brown University

Presentation title: Priority arguments

Providence, RI

Spring 2022

Contributed presentations

ASL North American Annual Meeting

Iowa State University

Presentation title: Computable categoricity relative to a c.e. degree

Ames, IA

Spring 2024

AMS New England Graduate Student Conference

Brown University

Presentation titles: Topology in the Reverse Math Zoo; Computable categoricity relative to a c.e. degree

Providence, RI

Spring 2024

17th International Conference on Computability, Complexity, and Randomness

Nagoya University

Presentation title: Computable categoricity relative to a c.e. degree

Nagoya, Japan

Spring 2024

Conferences and workshops attended

CoCoGEMS @TU Wien

Technische Universität Wien

Vienna, Austria

Summer 2026

CBMS Conference – Algorithmic Fractal Dimensions

Drake University

Des Moines, IA

Spring 2024

Computability and Combinatorics Summer School and Conference

UConn Hartford

Hartford, CT

Spring 2023

ASL Winter Meeting at the Joint Mathematics Meeting

Boston, MA

Spring 2023

IMS Graduate Summer School in Logic

National University of Singapore

Singapore

Summer 2022

Seminar presentations

Logic Workshop

CUNY Graduate Center

Presentation title: Switching categoricity behavior with a given degree

New York, NY

Fall 2026

Logic Seminar

George Washington University

Presentation title: Relativizing categoricity and low c.e. sets

Washington, D.C.

Spring 2026

Logic Seminar

University of Waterloo

Presentation title: Relativizing computable categoricity

Waterloo, ON

Fall 2025

SIGMA Seminar

University of Connecticut

Presentation title: The Scott Isomorphism Theorem

Storrs, CT

Spring 2025

Online Logic Seminar

Southern Illinois University

Presentation title: Computable categoricity relative to a degree

Online

Fall 2024

SIGMA Seminar

University of Connecticut

Presentation title: The Ginsburg–Sands theorem and computability theory

Storrs, CT

Spring 2024

SIGMA Seminar

Storrs, CT

<i>University of Connecticut</i>	<i>Spring 2024</i>
Presentation title: Normality and Randomness	
SIGMA Seminar	<i>Storrs, CT</i>
<i>University of Connecticut</i>	<i>Fall 2023</i>
Presentation title: Randomness and Hausdorff dimension	
Connecticut Logic Seminar	<i>Storrs, CT</i>
<i>University of Connecticut</i>	<i>Fall 2023</i>
Presentation title: Computable categoricity relative to a c.e. degree	
SIGMA Seminar	<i>Storrs, CT</i>
<i>University of Connecticut</i>	<i>Fall 2022</i>
Presentation title: When does the existence of an isomorphism imply the existence of a computable isomorphism?	

Grants and funding

Summer Doctoral Dissertation Fellowship	<i>Summer 2024</i>
<i>\$2,000 USD</i>	
Predoctoral Fellowship	<i>Spring 2024</i>
<i>\$7,805 USD</i>	

Outreach

President of the Association of Women in Mathematics	<i>Storrs, CT</i>
<i>University of Connecticut</i>	<i>2022-2024</i>
Speaker at the Mathematics Continued Conference	<i>Storrs, CT</i>
<i>University of Connecticut</i>	<i>Fall 2022</i>
The Mathematics Continued Conference seeks to give undergraduate students interested in math an opportunity to learn about graduate school and current research done by graduate students and faculty.	
Course Tutor for SSS Math Program	<i>Storrs, CT</i>
<i>University of Connecticut</i>	<i>Summer 2020</i>
Student Support Services (SSS) is a federally funded program at UConn which serves incoming students who are first-generation to college and/or come from communities underserved in higher education.	

Languages

English	<i>Second language learned,</i>
<i>Advanced proficiency in reading, writing, and speaking</i>	<i>learned in 2003</i>
Tagalog	<i>Native language</i>
<i>Intermediate proficiency in reading, writing, and speaking</i>	